

OS Lab

Practical-9

```
1. #include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>
#include <time.h>

int main() {
    int source_fd, dest_fd;
    char buffer[4096];
    ssize_t bytes_read, bytes_written;

    source_fd = open("input.txt", O_RDONLY);

    if (source_fd == -1) {
        perror("Error opening input file");
        return 1;
    }

    // Get file size using lseek()
    off_t file_size = lseek(source_fd, 0, SEEK_END);

    if (file_size == -1) {
        perror("lseek error");
        close(source_fd);
        return 1;
    }

    printf("File size: %ld bytes\n", file_size);

    // Move file pointer back to beginning
    lseek(source_fd, 0, SEEK_SET);

    dest_fd = open("output_lowlevel.txt",
        O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (dest_fd == -1) {
        perror("Error creating output file");
        close(source_fd);
    }
}
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    return 1;
}

clock_t start = clock();

while ((bytes_read = read(source_fd, buffer, sizeof(buffer))) > 0) {

    bytes_written = write(dest_fd, buffer, bytes_read);

    if (bytes_written != bytes_read) {
        perror("Error writing file");
        close(source_fd);
        close(dest_fd);
        return 1;
    }
}

clock_t end = clock();

close(source_fd);
close(dest_fd);

double time_taken =
    (double)(end - start) / CLOCKS_PER_SEC;

printf("File copied successfully using low-level I/O.\n");
printf("Time taken: %f seconds\n", time_taken);

return 0;
}
```

```

lahari@DESKTOP-AIC33F7:~$ cd ~/OSLab
lahari@DESKTOP-AIC33F7:~/OSLab$ ls
Lab      cr      lab.txt      pipes      section13_read.c  signal  signals
Test     file1.txt  memory      producer_consumer  section13pipescr.c  signal.c  skill4
cow      file_io   memory.c     producer_consumer.c  section13write     signal1  wr
cow.c    fork     named_createpipe  rd      section13write.c  signal1.c
lahari@DESKTOP-AIC33F7:~/OSLab$ mkdir file_io
mkdir: cannot create directory 'file_io': File exists
lahari@DESKTOP-AIC33F7:~/OSLab$ cd file_io
lahari@DESKTOP-AIC33F7:~/OSLab/file_io$ pwd
/home/lahari/OSLab/file_io
lahari@DESKTOP-AIC33F7:~/OSLab/file_io$ nano file_copy.c
lahari@DESKTOP-AIC33F7:~/OSLab/file_io$ gcc file_copy.c -o file_copy
lahari@DESKTOP-AIC33F7:~/OSLab/file_io$ ./file_copy
Error opening input file: No such file or directory
lahari@DESKTOP-AIC33F7:~/OSLab/file_io$ echo "Hello, this is a test file for Linux file I/O." > input.txt
lahari@DESKTOP-AIC33F7:~/OSLab/file_io$ ls
file_copy  file_copy.c  input.txt
lahari@DESKTOP-AIC33F7:~/OSLab/file_io$ cat input.txt
Hello, this is a test file for Linux file I/O.
lahari@DESKTOP-AIC33F7:~/OSLab/file_io$ ./file_copy
File size: 47 bytes
File copied successfully using low-level I/O.
Time taken: 0.000030 seconds
lahari@DESKTOP-AIC33F7:~/OSLab/file_io$ cat output_lowlevel.txt
Hello, this is a test file for Linux file I/O.
lahari@DESKTOP-AIC33F7:~/OSLab/file_io$ |

```

2. #include <stdio.h>

#include <stdlib.h>

#include <time.h>

int main() {

FILE *source, *destination;

char buffer[4096];

size_t bytes_read;

source = fopen("input.txt", "rb");

if (source == NULL) {

perror("Error opening input file");

return 1;

}

destination = fopen("output_library.txt", "wb");

if (destination == NULL) {

perror("Error creating output file");

fclose(source);

return 1;

}

clock_t start = clock();

```

while ((bytes_read = fread(buffer, 1, sizeof(buffer), source)) > 0) {

    if (fwrite(buffer, 1, bytes_read, destination) != bytes_read) {
        perror("Error writing file");
        fclose(source);
        fclose(destination);
        return 1;
    }
}

clock_t end = clock();

fclose(source);
fclose(destination);

double time_taken =
    (double)(end - start) / CLOCKS_PER_SEC;

printf("File copied successfully using standard library I/O.\n");
printf("Time taken: %f seconds\n", time_taken);

return 0;
}

```

```

lahari@DESKTOP-AIC33F7:~/OSLab/file_io$ nano file_copy.c
lahari@DESKTOP-AIC33F7:~/OSLab/file_io$ nano file_copy_library.c
lahari@DESKTOP-AIC33F7:~/OSLab/file_io$ gcc file_copy_library.c -o file_copy_library
lahari@DESKTOP-AIC33F7:~/OSLab/file_io$ ./file_copy_library
File copied successfully using standard library I/O.
Time taken: 0.000015 seconds
lahari@DESKTOP-AIC33F7:~/OSLab/file_io$ |

```

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3. #include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

```

```

int main() {
    int input_fd, output_fd;

    // Open input file
    input_fd = open("input.txt", O_RDONLY);

    if (input_fd == -1) {
        perror("Error opening input file");
        return 1;
    }
}

```

```

// Open/create output file
output_fd = open("redirected_output.txt",
                O_WRONLY | O_CREAT | O_TRUNC, 0644);

if (output_fd == -1) {
    perror("Error opening output file");
    close(input_fd);
    return 1;
}

// Redirect standard input (0)
if (dup2(input_fd, STDIN_FILENO) == -1) {
    perror("dup2 input error");
    return 1;
}

// Redirect standard output (1)
if (dup2(output_fd, STDOUT_FILENO) == -1) {
    perror("dup2 output error");
    return 1;
}

// Original file descriptors are no longer needed
close(input_fd);
close(output_fd);

printf("This output is redirected to redirected_output.txt\n");
printf("Standard output (stdout) is now connected to the file.\n");

return 0;
}

```

```

lahari@DESKTOP-AIC33F7:~/OSLab/file_io$ gcc io_redirect.c -o io_redirect
lahari@DESKTOP-AIC33F7:~/OSLab/file_io$ ./io_redirect
lahari@DESKTOP-AIC33F7:~/OSLab/file_io$ cat redirected_output.txt
This output is redirected to redirected_output.txt
Standard output (stdout) is now connected to the file.

```